# Exit Ticket · reflect and retrieve

## Exit Ticket

If a recursive call reaches its base case, all previous calls have already returned.

## Use key vocabulary from memory

1. Explain recursion in your own words.

2. Explain base case in your own words.

3. Explain recursive case in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 181 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Recursion: base cases and returns

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

A C# function is declared int Double(int n). What does return n * 2; do when n is 4?

### Q2

Starting total = 1, repeat total = total * i for i = 1, 2, 3. What is total after each iteration?

### Q3

Values A, B and C are pushed onto a stack in that order. Which leaves first when popping?

### Q4

A function AddTwo(x) returns x + 2. Calculate AddTwo(AddTwo(3)) and show the inner and outer calls.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Start with a small input — Factorial(4) calls Factorial(3); it cannot finish multiplying until the inner call returns.

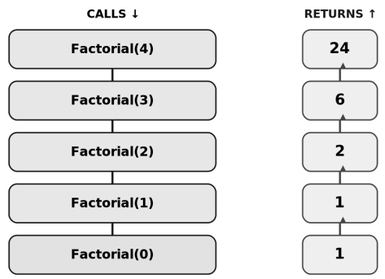
## Core idea 2

Reach the stopping rule — Factorial(0) returns 1. It makes no further recursive call.

## Core idea 3

Return through the calls — Factorial(1) = 1 × 1; Factorial(2) = 2 × 1; Factorial(3) = 3 × 2; Factorial(4) = 4 × 6 = 24.

**Key visual · complete concept**








# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Modify Factorial to reject negative input; write the C# guard and explain the reason.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

### Question 1

Complete the calls and returns for Factorial(2).

### Question 2

What happens if Factorial has no base case? Explain using the call stack.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

### Vocabulary · use the correct technical terms

recursion — A function calling itself to solve a smaller problem.

base case — A condition that stops further recursive calls.

recursive case — A call to the same function with a smaller or simpler input.

call stack — Memory tracking active function calls and their local state.

### Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

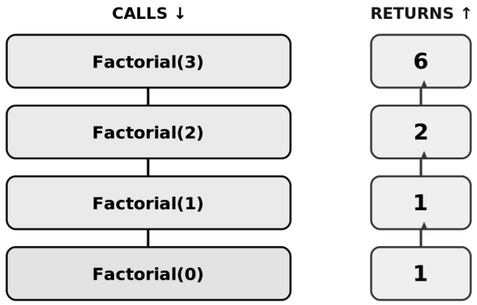
## Problem

Trace Factorial(3). Show each call and its return value.

## Follow the worked method

1. Calls — 3 → 2 → 1 → 0. The argument decreases by one each time.
2. Base case — Factorial(0) returns 1. No more calls occur.
3. Returns — Factorial(1)=1, Factorial(2)=2, Factorial(3)=6. Final answer: 6.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict the next call
4 calls 3; 3 calls 2; 2 calls 1; 1 calls 0.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.